

ics products like digital TVs (sink devices), set-top boxes, BD/DVD players, media players, tablets, mobile phones (all source devices),

automobiles, car infosystems and so on. HDMI uses transition minimised differential signaling, which enables you to enjoy high-definition

video and digital audio content. Ideally, HDMI supports all standard-definition, high-definition and ultra-high-definition videos in a TV, different colour space, deep colour, video aspect ratio, Dolby, DTS audio and so on.

There are various versions of HDMI, out of which HDMI 1.4b is the most common. It is available for almost all consumer electronics products that support HDMI. HDMI's first version was HDMI 1.0 and the latest released and approved is HDMI 2.0a. More information can be found on www.hdmi.org.

HDMI can be of two types: wired and wireless. Table I shows HDMI support with different versions.

To certify that a product supporting HDMI is abiding with HDMI standards, the product must have HDMI logo on it. If the product does not have HDMI logo, it may or may not work well with different HDMI-connected devices. The logo gives assurance that the product is HDMI-compliant and certified.

HDMI also enables you to control various source devices connected to a sink (such as TV) via a single remote control unit. This feature is known as consumer electronic control (CEC). Various companies name it differently as listed below:

Samsung: Anynet +
Sharp: Aquos Link
Sony: Bravia Sync
Hitachi: HDMI - CEC
AOC: E-Link
Pioneer: Kuro Link
Toshiba: Regza Link
Philips: Easy Link
Panasonic: Viera Link

To display the content and play audio from HDMI interface, the source and sink device follows certain protocols, which are:

- EDID
- HDCP
- Electrical
- Packet protocol
- CEC/ARC

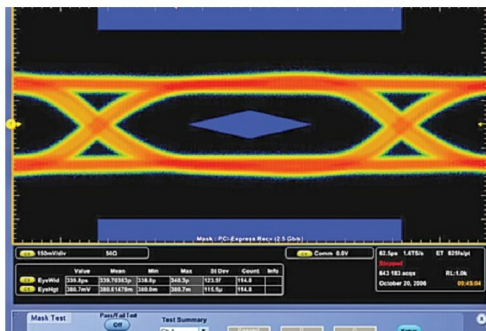


Fig. 2: Electrical characteristics during transition minimised differential signaling

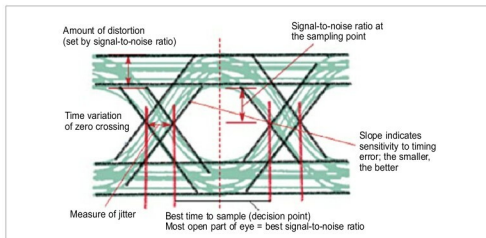


Fig. 3: Interpreting an eye diagram

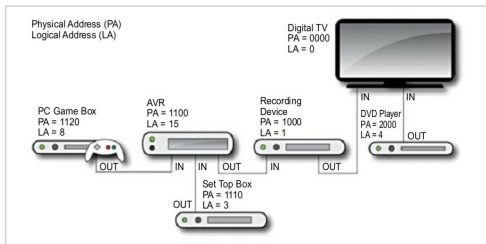


Fig. 4: CEC network